

TEL-TSI

Introduction to semantic modelling

A **concept scheme** (in SKOS: *ConceptScheme*) is a structured collection of concepts that carry a fixed, shared meaning within a particular domain. The goal is for everyone working with the model — people and systems alike — to use the same words for the same things. That sounds obvious, but in practice language causes a surprising amount of confusion: what one organisation calls a *measurement point*, another calls a *sensor*, a *location*, or an *observation station*.

A concept scheme solves this by recording, for each concept:

- a **preferred label** (prefLabel) it is known by,
- a **definition** that unambiguously describes what the concept does and does not cover,
- and optionally a **scope note** or **example** to clarify borderline cases.

Concepts rarely stand on their own. They are related **hierarchically** (a vehicle is broader than a truck) or **associatively** (a road user relates to a road segment). These relationships make it possible for queries and analyses to navigate between general and specific concepts, and to make connections between concepts explicit. Within SKOS we use `broader`, `narrower`, and `related` for that.

When a concept also exists elsewhere — in a national or European standard, for instance — we record that as a **mapping relation**. This prevents reinventing the wheel: we link to the source so interoperability with external systems remains possible.

This document describes the concept scheme as it is currently recorded. The following sections list the metadata, every concept, the relationships between them, and — where the model is connected to a data source — how it is linked to that source data.

TEL-TSI

Description

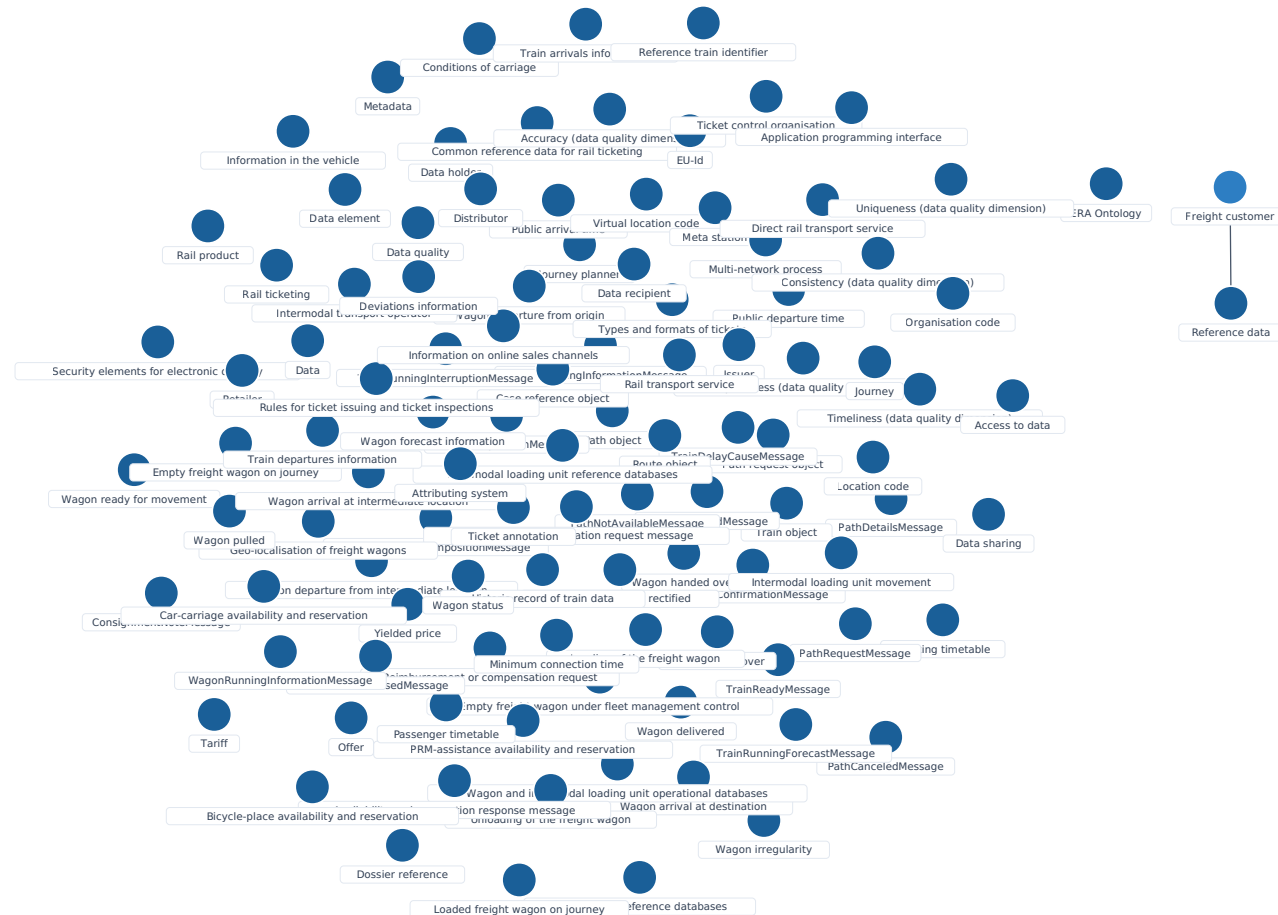
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Metadata

Field	Value
URI	https://github.com/NAPCORE/its-data-dictionary/tel-tsi
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Organization	NAPCORE
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Model overview

Visual overview of the concepts and their relations. Hierarchical relations are drawn with solid arrows; associative relations are drawn with dashed arrows.



→ broader / narrower (hiërarchisch / Associatief)
 - - - - - related (gerelateerd)

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Concepts

Concepts overview

Concept	Definition
Reference train identifier	Object that uniquely identifies a planned train transport (PlannedTransportIdentifier, type TR), specified by the lead railway undertaking, used as the reference for a train across networks.
Train object	Object (TrainID) identifying a train, specified by the lead railway undertaking, to which operational train running data is related.
Path request object	Object (PathRequestID) identifying a request for infrastructure capacity, specified by the lead railway undertaking; carries the operational train number and commercial train number as attributes.
Path object	Object (PlannedTransportIdentifier, type PA) identifying an allocated train path, specified by the infrastructure manager.
Route object	Object (PlannedTransportIdentifier, type RO) identifying a route, specified by the lead railway undertaking.
Case reference object	Object (PlannedTransportIdentifier, type CR) identifying a case reference, specified by the holder of the data.
PathRequestMessage	Message by which an applicant requests infrastructure capacity (a train path) from the infrastructure manager, in accordance with the ERA Ontology.
PathDetailsMessage	Message by which the infrastructure manager provides the details of a path in response to a path request, in accordance with the ERA Ontology.
PathConfirmedMessage	Message by which the applicant confirms the path details proposed by the infrastructure manager.
PathDetailsRefusedMessage	Message by which the applicant refuses the path details proposed by the infrastructure manager.
PathCanceledMessage	Message by which a previously confirmed or requested path is cancelled.
PathNotAvailableMessage	Message by which the infrastructure manager notifies that a requested path is not available.
ReceiptConfirmationMessage	Message confirming receipt of a message between telematics stakeholders.

Concept	Definition
TrainCompositionMessage	Message conveying the composition of a train, including vehicle-level information, total train mass, European Vehicle Number of traction units, and flags for dangerous goods, intermodal loading units, armed forces and quieter routes.
PassengerTrainCompositionMessage	Message conveying the composition of a passenger train, including seating maps, classes, coaches for persons with reduced mobility and bicycle spaces.
TrainReadyMessage	Message by which the railway undertaking notifies that a train is ready for departure; status values include 'NotReady' with sub-fields for train delay, train ready time and delay cause.
TrainRunningInformationMessage	Message conveying actual train running information at reporting points.
TrainRunningForecastMessage	Message conveying forecast train running information, with expected accuracy as set out in the regulation's accuracy methodology.
TrainDelayCauseMessage	Message conveying the cause of a train delay.
TrainRunningInterruptionMessage	Message notifying an interruption to the running of a train.
Working timetable	The working timetable data as defined in Article 3(28) of Directive 2012/34/EU, published as open data under a Creative Commons BY-ND 4.0 licence.
Historic record of train data	Retained record (at least 12 months) of train data, containing the working timetable, reference train identifier, commercial train number, reporting locations, actual times, delay and delay cause, and train composition.
ConsignmentNoteMessage	Message conveying the electronic consignment note (eCN), i.e. the transport document referred to in Article 6 of Council Regulation No 11/1960, sealed with a qualified electronic seal pursuant to Regulation (EU) No 910/2014.
WagonRunningInformationMessage	Message conveying running information about a freight wagon across its movement life cycle; decomposed into the individual wagon events specified in the Annex.
Intermodal loading unit movement	Data conveying the movement of an intermodal loading unit.
Wagon forecast information	Forecast information for a freight wagon, with accuracy determined by the regulation's accuracy methodology.
Geo-localisation of freight wagons	Wagon event conveying the geographic position of a freight wagon.

Concept	Definition
Wagon status	Wagon event conveying the current status of a freight wagon.
Wagon ready for movement	Wagon event indicating that a freight wagon is ready to be moved.
Wagon pulled	Wagon event indicating that a freight wagon has been pulled.
Wagon departure from origin	Wagon event indicating departure of a freight wagon from its origin.
Wagon arrival at intermediate location	Wagon event indicating arrival of a freight wagon at an intermediate location.
Wagon departure from intermediate location	Wagon event indicating departure of a freight wagon from an intermediate location.
Wagon handed over	Wagon event indicating that a freight wagon has been handed over between responsible parties.
Wagon taken over	Wagon event indicating that a freight wagon has been taken over by a responsible party.
Wagon irregularity	Wagon event reporting an irregularity affecting a freight wagon.
Wagon irregularity rectified	Wagon event indicating that a previously reported wagon irregularity has been rectified.
Wagon arrival at destination	Wagon event indicating arrival of a freight wagon at its destination.
Wagon delivered	Wagon event indicating that a freight wagon has been delivered.
Rolling stock reference databases	Reference databases for rolling stock, including the European Vehicle Number, keeper, type and technical parameters; managed by the rail sector and linked from the ERA common central repository.
Intermodal loading unit reference databases	Reference databases for intermodal loading units, including ILU code, type, dimensions and owner.
Wagon and intermodal loading unit operational databases	Operational databases recording the operational states of freight wagons and intermodal loading units across their movement.
Loading of the freight wagon	Operational state indicating that a freight wagon is being loaded.

Concept	Definition
Loaded freight wagon on journey	Operational state indicating that a loaded freight wagon is on its journey.
Empty freight wagon on journey	Operational state indicating that an empty freight wagon is on its journey.
Unloading of the freight wagon	Operational state indicating that a freight wagon is being unloaded.
Empty freight wagon under fleet management control	Operational state indicating that an empty freight wagon is under fleet management control.
Dossier reference	Reference identifying a ticketing dossier, used to link tickets, reservations and after-sale processes.
Availability and reservation request message	Message by which availability is checked and a reservation is requested for a rail product.
Availability and reservation response message	Message conveying the response to an availability and reservation request.
PRM-assistance availability and reservation	Message for checking availability of and reserving assistance for persons with disabilities or reduced mobility.
Bicycle-place availability and reservation	Message for checking availability of and reserving a bicycle place.
Car-carriage availability and reservation	Message for checking availability of and reserving the carriage of cars, motorcycles or boats.
Ticket annotation	After-sale message recording an annotation on a ticket.
Reimbursement or compensation request	After-sale message requesting reimbursement or compensation in respect of a ticket.
Passenger timetable	The date and time of passenger rail transport services operated as a commercial service by a railway undertaking during a given time interval, used in the context of rail ticketing and rail passenger travel information referred to in Section 4 of the Annex.
Minimum connection time	The minimum amount of time that is sufficient for a passenger to connect between the access nodes of two consecutive passenger transport services included in a journey.
Tariff	Aggregated data established by a railway undertaking (or data holder) providing all relevant tariff details, including the rail-transport-service price due by a distributor to the railway undertaking and the associated attributes and conditions.

Concept	Definition
Offer	Aggregated data established by a distributor combining applicable tariffs, with the rail-product price due by the retailer to the distributor.
Yielded price	Price associated with a tariff resulting from an algorithm based on variables, referred to as the yielding rule.
Rules for ticket issuing and ticket inspections	Data establishing the rules governing the issuing of tickets and the inspection of tickets.
Information on online sales channels	Data describing the online sales channels through which rail products are offered.
Conditions of carriage	Data establishing the conditions of carriage, decomposed for passengers, persons with reduced mobility (with rolling-stock and station accessibility), luggage, bicycles, and cars/motorcycles/boats.
Types and formats of tickets	The catalogue of ticket types and formats, including secured-paper, blank-paper and mobile/flexible-size ticket layouts with their associated security models.
Common reference data for rail ticketing	Code lists for ticketing managed and assigned by the Agency, used as a basis for data sharing in the rail ticketing domain.
Security elements for electronic delivery	Security elements used for the electronic delivery of tickets.
Train departures information	Data on train departures for display at stations.
Train arrivals information	Data on train arrivals for display at stations.
Deviations information	Data on deviations from the scheduled service for display at stations.
Information in the vehicle	Passenger travel information provided inside the vehicle.
Distributor	Entity with the technical capacity to combine rail products from one or more railway undertakings to supply an issuer or retailer, with rights to operate without an initial query to attributing systems and autonomous rights to check availability of price.
Issuer	Entity with the legal and technical capacity to issue, for a retailer, a ticket on which the parties to the contract of carriage are indicated.
Retailer	

Concept	Definition
	Entity that combines and/or sells rail products from one or more distributors, obtaining access via an initial request; it may apply fees or reductions but cannot amend rail-product content.
Attributing system	The railway undertaking's digital system hosting the online catalogue of individual rail transport services and enabling distributors to check availability and make reservations.
Journey planner	A search engine used to find an optimal journey based on defined criteria set out by its user.
Ticket control organisation	An organisation empowered to inspect passenger tickets, including at platform access doors.
Rail product	A tailored service in relation to a passenger rail transport service linked to a tariff.
Meta station	A group of stations within a functional urban area, including one or more urban nodes listed in Annex II to Regulation (EU) 2024/1679, that may extend across multiple Member States and where distinct stations can be reached by connecting urban transport modes and urban transport services, whether scheduled or unscheduled.
Data	A machine-readable representation of acts, facts or information and any compilation of such acts, facts or information intended to be shared in a harmonised and structured format.
Data sharing	The provision of data by a data holder to a data recipient for the purpose of the joint or individual use of such data, directly or through an intermediary, for example under open or commercial licences subject to a fee or free of charge.
Rail ticketing	The process comprising at least one of the following for passenger rail transport services: the digital presentation of services for journey planning; the digital checking of availability or reservation of rail products for sales; the digital issuing and control of tickets; the digital handling of after-sale processes.
Rail transport service	A scheduled service for carriage by rail between a point of departure and a point of arrival, covering transport services for goods, passengers, luggage, bicycles or cars, and tailored transport arrangements, as well as passenger assistance including services for re-routing and journey continuation.
Access to data	The right to consult the data with or without control over it, distinctively from other rights and obligations relating to its use for non-commercial or commercial purposes.
Multi-network process	

Concept	Definition
	A process that is affected by the operation of a rail transport service across two or more networks managed by different infrastructure managers or operators of rail freight service facilities.
Direct rail transport service	A rail transport service that may include intermediate commercial stops between its origin and destination but does not involve any transfer of passengers or freight between trains.
Application programming interface	A set of functions, procedures, definitions and protocols for interoperable data sharing between telematics applications.
Journey	An end-to-end trip of a passenger or a load that can include multiple legs.
Public departure time	The boarding time limit for passengers, that is the point in time when access to the train, platform or station is closed in preparation for departure.
Public arrival time	The alighting time for passengers, that is the point in time when the doors of the train are opened at the destination platform or terminal.
Data holder	A natural or legal person that has the right to use or an obligation to produce and make available data.
Data recipient	A natural or legal person that has the right to access the data produced and made available by the data holder under specified access conditions for use.
Intermodal transport operator	An entity that organises freight transport services connecting two or more freight terminals with more than one transport mode.
Freight customer	The entity that has issued consignment information to the freight railway undertaking and has concluded a contract of carriage; may be a logistics service provider, intermodal transport operator, freight forwarder, fleet manager of empty wagons, or the data holder.
ERA Ontology	A technical document issued by the Agency pursuant to Article 4(8) of Directive (EU) 2016/797, setting out human-readable and machine-readable data definitions and presentations and the associated quality and accuracy requirements for data elements of the Union rail system.
Data element	A field, value, or attribute of the data shared, structured consistently with the overall protocol used for data sharing and carrying essential information that contributes to the meaning and purpose of the data shared, enabling data sharing between systems, applications or organisations.
Reference data	A data set of standardised and interoperable single identifiers to be used as a basis for data sharing in a specific domain.

Concept	Definition
Metadata	Metadata within the meaning of Article 2(2) of Regulation (EU) 2023/2854 (Data Act).
Data quality	The degree of fitness for use at which the characteristics of data satisfy the stated and implied level of accuracy, completeness, consistency, timeliness and uniqueness of datasets, and create actionable insights for other users under specified conditions.
Accuracy (data quality dimension)	Data quality dimension expressing the degree to which data correctly represents the real-world value it is intended to represent.
Completeness (data quality dimension)	Data quality dimension expressing the degree to which all required data is present.
Consistency (data quality dimension)	Data quality dimension expressing the degree to which data is free of contradiction across datasets and over time.
Timeliness (data quality dimension)	Data quality dimension expressing the degree to which data is available within the required time.
Uniqueness (data quality dimension)	Data quality dimension expressing the degree to which there is no unwanted duplication of data.
Organisation code	Single identifier of any organisation or role involved in rail data sharing, assigned by the Agency.
Location code	Single identifier of every geographical point relevant to rail data sharing, assigned by the Agency.
Virtual location code	Code assigned to non-rail-network locations used by rail-replacement transport services, services sold as rail but not operated on rail, and meta stations.
EU-Id	The canonical European identifier used by data holders when feeding the national access points.